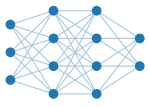
Physics-informed training makes NMPC more robust, not more accurate

1 Two training objectives

identical 3 × 64 *network*

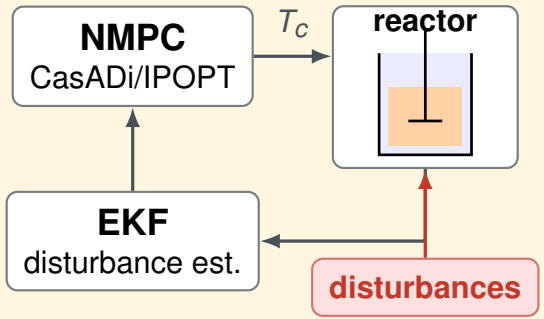


Black-box NN
data loss only

PINN
data loss
+ **physics loss**

2 Same offset-free NMPC

open-loop-unstable CSTR



black-box · PINN · first-principles model
every result repeated over 7 seeds

3 Result (7 seeds)

Runaways (112 sweep runs)

−70%, **fewer in all 7 seeds**

Feed-temperature disturbance
error **9.83** → **6.37 K** (p = 0.033)

Nominal: no significant gain
Solver: faster MPC solves